

Module 8: Normal Distribution

Background

Recall with probability distributions:

- The outcomes are random events.
- All outcomes are assigned a probability.
- The probabilities are numbers between 0 and 1. This makes sense because each probability is a relative frequency.
- The sum of all of the probabilities is 1. This makes sense because we have listed all the outcomes. Since each probability is a relative frequency, these outcomes make up 100% of the observations.

Random variables can be either **discrete** or **continuous**.

Binomial Distribution... Example of a discrete distribution.

Continuous variables (decimal values), such as height of adults and body temperature of rats, can assume all values between any two given values of the variables.

In a continuous probability distribution, probabilities are represented as areas under a curve.

The **normal distribution** can be used to describe many variables since the variations from a normal distribution are small.

Normal Distributions

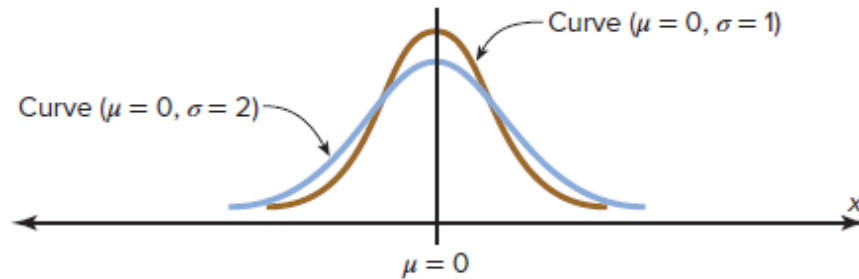
A **normal distribution** is a mathematical model with a smooth bell-shaped curve to describe the bell-shaped data distributions. A normal distribution has the following characteristics:

1. A continuous random variable.
2. Symmetrical around the mean, μ , the left side is a mirror image of the right side centered at the mean.
3. There is one peak at the mean of bell-shaped distributions.

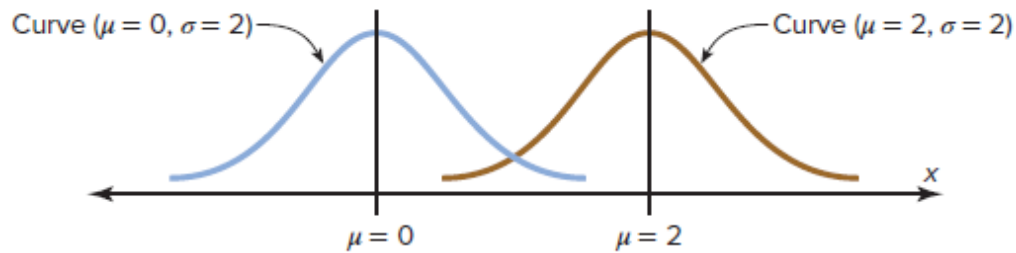
Module 8: Normal Distribution

The shape and position of the normal distribution curve depend on the **mean** and **standard deviation**.

The larger the standard deviation the wider the curve.



(a) Same means but different standard deviations

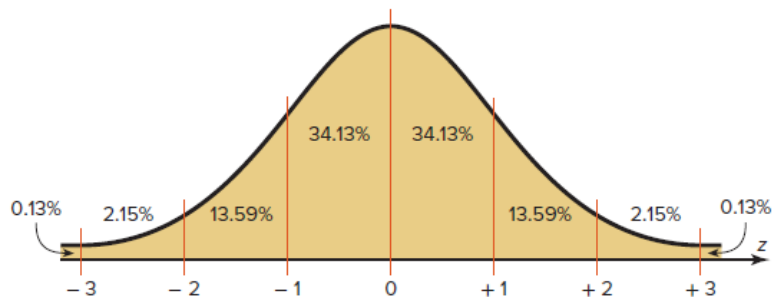


(b) Different means but same standard deviations

When the population means are the same, but the population standard deviations are different, the curve with the larger population standard deviation is more **spread out**.

When the population standard deviations are the same, but the means are different, the curves have the **shape** and **spread**, but are **centered** at different points along the x axis.

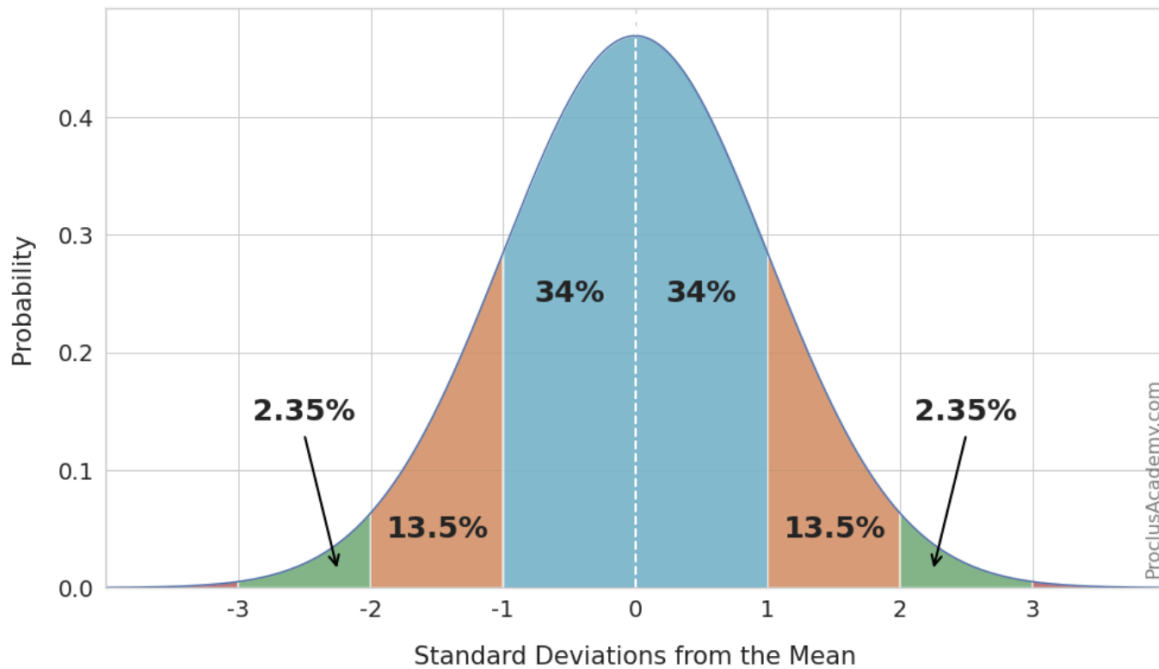
The area under the normal distribution follows the **Empirical Rule**:



Module 8: Normal Distribution

Standard Normal Distribution

The Standard Normal Distribution is a normal distribution with a $\mu = 0$ and a $\sigma = 1$. Always these values



The formula for the standard normal distribution is $y = \frac{e^{-x^2/2}}{\sqrt{2\pi}}$.

Need Z-score for this.

All normally distributed variables can be transformed into the **standard normal curve** by using the formula for the **Z- score**:

value is random variable

$$Z = \frac{\text{value} - \text{mean}}{\text{standard deviation}} \quad \text{or} \quad Z = \frac{X - \mu}{\sigma}.$$

Calculator always gives the area to the left on the bellcurve.

The standard score, or z-score, is the number of **standard deviations** that a particular X value is from the mean.

Note:

- The cumulative area for $z = 0$ is 0.5000.
- The cumulative area is close to 0 for z-scores close to $z = -3.49$.
- The cumulative area is close to 1 for z-scores close to $z = 3.49$.

You can use tables or technology to find the corresponding areas under the standard normal distribution curve.

Module 8: Normal Distribution

Finding Areas Under the Standard Normal Distribution Curve

Step 1: Draw the normal distribution curve, label the Z-score, and shade the area under consideration.

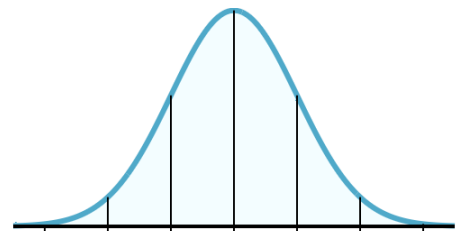
Step 2: Choose the appropriate type of problem and either use Table E in Appendix A or use technology to find the shaded area.

Area:	Calculation:
The area under the standard normal distribution curve to the left of a z value	Find the z value in Table E and use the area given.
The area under the standard normal distribution curve to the right of a z value	Find the z value in Table E and subtract the area given from 1.
The area under the standard normal distribution curve between two z values	Find both z values in Table E and subtract the corresponding areas.

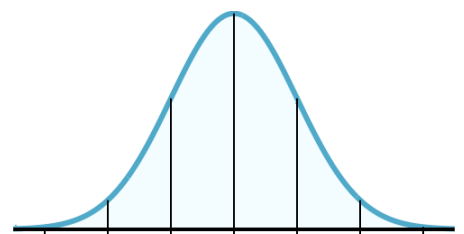
Example:

Find the area under the standard normal distribution curve

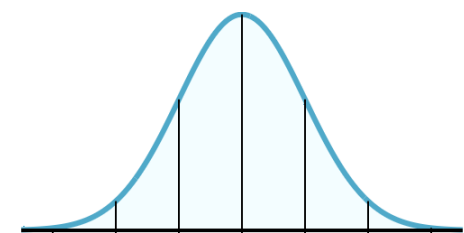
- a. To the left of $z = -1.03$.



- b. To the right of $z = 1.23$

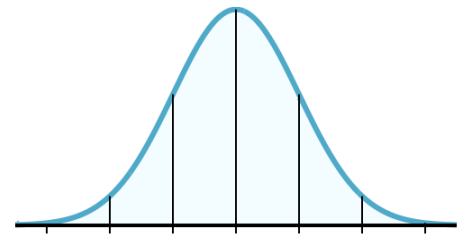


- c. Between $z = 0$ and $z = -0.32$.



Module 8: Normal Distribution

- d. Between $z = -2.15$ and $z = .97$.

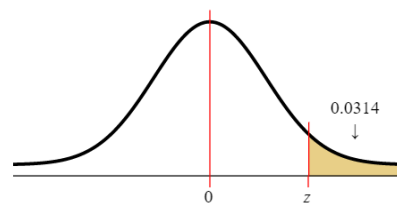
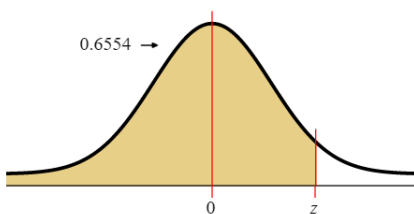


Example: Find the Probabilities using the standard normal distribution curve

a. $P(z > 1.12)$

b. $P(-1.20 < z < 1.56)$

Example: Find the z-value that corresponds to the given area in the figure below.



1.86

Module 8: Normal Distribution

Applications of the Normal Distribution

The standard normal distribution curve can be used to solve many practical problems. Assume the variables presented here are all **normally** or **approximately normally distributed**.

First we transform the original variable to a standard normal distribution variable using the formula,

$$z = \frac{\text{value} - \text{mean}}{\text{standard deviation}} \text{ or } z = \frac{X - \mu}{\sigma}, \text{ rounding the result to two decimal places.}$$

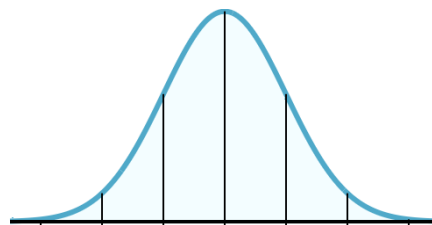
To Find the Area Under Any Normal Curve

Step 1	Draw a normal curve and shade the desired area.
Step 2	Convert the value of X to z using the formula.
Step 3	Find the area under the curve using a table, calculator or software.

Example: Liters of Blood

An adult has on an average 5.2 liters of blood. Assume the variable is normally distributed and has a standard deviation of 0.3. Find the percentage of people who have more than 5.4 liters of blood in their system.

Step 1: Draw a normal curve and **shade** the desired area.



Step 2: Use the formula to find the z value corresponding to 5.4.

$$z = \frac{X - \mu}{\sigma}$$

Step 3: Use tables, your calculator, or computer software to determine the area to the right of z . (Notice this value is rounded.) The area under the standard normal curve to the left of the z -score, then to find the area to the right, subtract the value from the table from 1.

Therefore, 25.14 % of adults have greater than 5.4 liters of blood in their system.

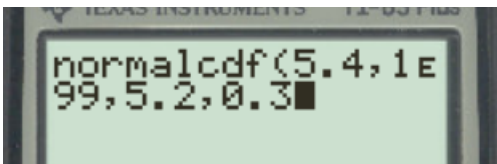
Module 8: Normal Distribution

To calculate the above answer using a program in your TI-calculator:

1. Press 2^{nd} , VARS to access the distribution list.
2. Select normalcdf(
3. Type in these 4 values: Lower limit, Upper limit, mean, standard deviation. Then press enter.

Note: If the area is only bounded on one side, you'll use "infinity" or "negative infinity" as the bound. We can't type in infinity, so we'll type a large number (in scientific notation.) For area in the right tail, use $1E99$, and for area in the left tail, use $-1E99$. (This E is the scientific notation button and is found by pressing 2^{nd} , comma. It looks like EE on the button but shows up as one E on your screen.)

Example: For the previous example, type normalcdf(5.4,1E99,5.2, 0.3).



Example: Monthly Mortgage Payments

The average monthly mortgage payment including principal and interest is \$982 in the United States. If the standard deviation is approximately \$180 and the mortgage payments are approximately normally distributed, find the probability that a randomly selected monthly payment is a) more than \$1000; b) More than \$1475; c) between \$800 and \$1150?

a)

b)

Module 8: Normal Distribution

c)

Formula for Finding the Value of a Normal Variable X

$$X = z \cdot \sigma + \mu$$

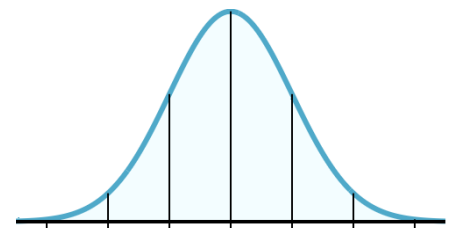
Finding Data Values for Specific Probabilities

Step 1	Draw a normal curve and shade the desired area that represents the probability, proportion or percentile.
Step 2	Find the z value corresponding to the desired area from the table, calculator, or computer software.
Step 3	Calculate the X value using the formula $X = z \cdot \sigma + \mu$.

Example: Police Academy Qualifications

To qualify for a police academy, candidates must score in the top 10% on a general abilities test. Assume the test scores are normally distributed and the test has a mean of 200 and a standard deviation of 20. Find the lowest possible score to qualify.

Step 1. Draw a normal distribution curve and shade the right hand 10% of the area.



Step 2. The area to the left of 0.1 is $1 - 0.1 = 0.9$. Find the z value from corresponding to 0.9 area to the left.

Step 3. Use this value in the formula $X = z \cdot \sigma + \mu$ to find X .

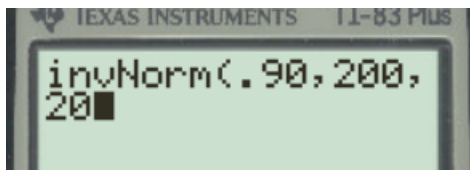
Module 8: Normal Distribution

Conclusion: A score of _____ should be used for the cutoff. Anyone scoring _____ or _____ qualifies for the academy.

To calculate the above answer using a program in your TI-calculator:

1. Press 2nd, VARS to access the distribution list.
2. Select invNorm(
3. Type in these 4 values: Area to the left, mean, standard deviation. Then press enter.

Example: For the previous example, type invNorm(.90,200,20).

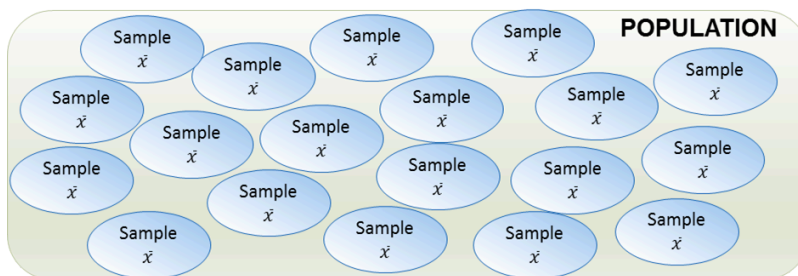


The Central Limit Theorem

Select a sample of 30 adult males and find the mean measure of the triglyceride levels for the sample to be 187 mg/dl. Select a second sample, and find the mean to be 192 mg/dl. Continue selecting until you have 100 samples and sample means. Let the mean of these sample means be a random variable. The sample means becomes a *sampling distribution of sample means*.

Definition: Sampling Distribution of Sample

A **sampling distribution of sample means** is a distribution using the means computed from all possible random samples of a specific size taken from a population.



Module 8: Normal Distribution

The sample means of samples randomly selected with replacement will be somewhat **different** from the population **mean**. The differences are caused by **sampling error**.

Definition: Sampling Error

Sampling Error is the difference between the sample measure and the corresponding population measure due to the fact that the sample is not a perfect representation of the population.

When all possible samples of a specific size are selected with replacement from a population, the distribution of the sample means for a variable has two important properties:

Properties of the Distribution of Sample Means
1. The mean of the sample means will be the same as the population mean.
2. The standard deviation of the sample means will be smaller than the standard deviation of the population, and it will be equal to the population standard deviation divided by the square root of the sample size.

The Central Limit Theorem

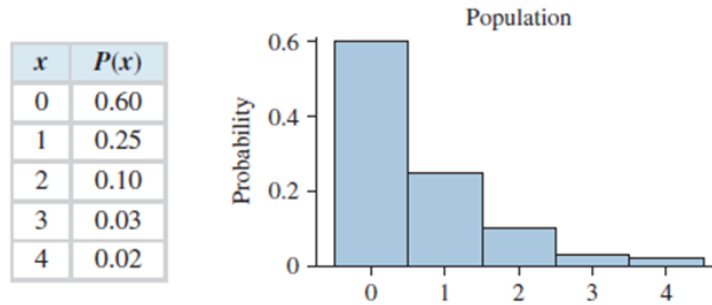
As the sample size n increases without limit, the shape of the distribution of the sample means taken with replacement from a population with mean μ and standard deviation σ will approach a normal distribution with a mean of μ and a standard deviation of $\frac{\sigma}{\sqrt{n}}$.

For samples with sufficient size, the central limit theorem can be used to answer questions about sample means in the same manner that a normal distribution can be used to answer questions about individual values. The difference is that the formula for z becomes $z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$

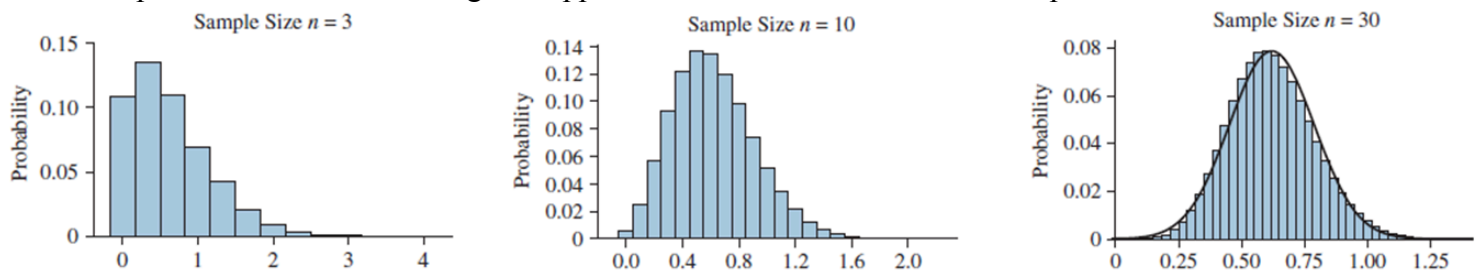
1. When the original variable is normally distributed, the distribution of the sample means will be normally distributed for any sample size n .
2. When the distribution of the original variable is not normal, a sample size of 30 or more is needed to use a normal distribution to approximate the distribution of the sample means.

Module 8: Normal Distribution

If a population is skewed, then a larger sample size is necessary for the sampling distribution of $\bar{x}$ to be approximately normal. Consider the following probability distribution.



Below are the probability histograms for the sampling distribution of $\bar{x}$ for samples of size 3, 10, and 30. Note that the shapes of the distributions begin to approximate a normal curve as the sample size increases.



Example: For which of the following scenarios can you use the normal curve to calculate probabilities? What is the mean and standard deviation for each?

- A sample of size 45 will be drawn from a population with mean 15 and standard deviation 3.5.
- A sample of size 8 will be drawn from a population with mean 60 and standard deviation 5.
- A sample of size 24 will be drawn from a population with mean 35 and a standard deviation of 1.2.

Module 8: Normal Distribution

Example: Working Weekends.

The average time spent by construction workers who work on weekends is 7.93 hours (over a period of 2 days).

Assume the distribution is approximately normal and has a standard deviation of 0.8 hours.

- a. Find the probability that an individual who works at that trade works fewer than 8 hours on the weekend.

- b. If a sample of 40 construction workers is randomly selected, find the probability that the mean of the sample will be less than 8 hours.

Example: The standard deviation of a variable is 13. If a sample of 65 individuals is selected:

- a. Find the standard error of the mean.

- b. Find the sample size needed to cut the standard error of the mean in half.